The first case past the boundary: $p = 2$, $G = \mathbb{Z}/4$, $d = 3$

the class survives — and the Witt carry, killed by the bound, is the whole story

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Result. For $p = 2$, $G = \mathbb{Z}/4$ ($r = 2$), $d = 3$ — the smallest genuine case with $r < d$ — the symmetric power $\mathcal{P}^{[3]_G}$ is *unramified* at η_C. Equivalently the top Artin–Schreier class survives the base change of the inductive step; there is no absorption; **?? holds for** $(r, d) = (2, 3)$. The computation is short, completely explicit, and verified symbolically in the companion `wittcheck.py`. It also pinpoints the mechanism: the only obstruction is a single *Witt carry* term, and the ramification bound is exactly the condition that strips its pole.

1 Why $p = 2$ is the only $(r, d) = (2, 3)$ test

For a totally ramified $\mathbb{Z}/p^2$ Artin–Schreier–Witt extension with bottom datum of pole order m_0 (prime to p) and top datum of pole order m_1, the upper breaks are $u_1 = m_0$ and $u_2 = \max(p\, m_0,\, m_1)$. The bound "ram$(\mathcal{P}) \le d$" is $u_2 < d$ (**??**, $G^d = 1$). At $d = 3$:

$$u_2 < 3 \text{ with } m_0 \ge 1 \implies p\, m_0 < 3 \implies p < 3 \implies p = 2.$$

For $p \ge 3$ the bottom layer would have to be *unramified* ($m_0 = 0$), degenerating to the proven $\mathbb{Z}/p$ base case. So the unique genuinely-wild $(r, d) = (2, 3)$ instance is $p = 2$, $m_0 = 1$ (hence $u_1 = 1$, $u_2 = 2 < 3$), and any $m_1 \le 2$. We treat it.

2 The symmetric-power Witt datum

Work in the local model of the proven $\mathbb{Z}/p$ computation (`RamificationAfterBlowupIsTame.tex`, eq. **??**): at the blowup point V_C on $C^{(3)}$ the completed field is $\widehat{K_C} = \kappa_C((e_3))$, with $e_3 = x_1x_2x_3$ a uniformiser, $\kappa_C = k\big(e_1/e_3,\, e_2/e_3\big)$, and the valuation rule

$$v_C(e_i) = 1 \ (i = 1, 2, 3), \qquad v_C(e_i/e_j) = 0. \tag{1}$$

Locally $\mathcal{P}$ is the W_2-Artin–Schreier–Witt torsor $\wp(\vec{Y}) = \vec{a}$, $\vec{a} = (a_0, a_1)$ over $k((x))$, with $a_0 = x^{-1}$ (so $m_0 = 1$) and a_1 of pole $m_1 \le 2$. By the same Ξ-invariant computation as the $\mathbb{Z}/p$ case

(`RamificationAfterBlowupIsTame.tex`, lines 521–547), the symmetric power $\mathcal{P}^{[3]_G}$ is the W_2-torsor over $\widehat{K_C}$ with datum the *Witt sum*

$$\vec{\alpha} \;=\; {\textstyle\sum_{i=1}^{3}}^{W}\, \vec{a}(x_i) \;=\; (\alpha_0, \alpha_1).$$

In characteristic 2 the length-two Witt carry is $C(a,b) = \frac{a^2+b^2-(a+b)^2}{2} = ab$, and folding over the three points gives (verified in `wittcheck.py`, step B)

$$\alpha_0 = \sum_i a_0(x_i), \qquad \alpha_1 = \underbrace{\sum_i a_1(x_i)}_{\text{top sum}} + \underbrace{e_2(s_1,s_2,s_3)}_{\text{Witt carry}}, \qquad s_i := a_0(x_i) = x_i^{-1}. \tag{2}$$

The carry $e_2(s) = \sum_{i<j} s_i s_j$ is the only new feature beyond the $\mathbb{Z}/p$ case; it is the entire content of the second Witt layer.

3 The carry is integral exactly when the bound holds

Both pieces of $\vec{\alpha}$ are now read off at V_C via (1) (`wittcheck.py`, step C):

$$\alpha_0 = \sum_i x_i^{-1} = \frac{e_2}{e_3} \;\; (v_C = 0); \qquad e_2(s) = \sum_{i<j} \frac{1}{x_i x_j} = \frac{e_1}{e_3} \;\; (v_C = 0).$$

Using $e_k(x^{-1}) = e_{d-k}(x)/e_d(x)$, the carry for general d and m_0 is

$$e_2\big(x_1^{-m_0}, \ldots, x_d^{-m_0}\big) = \text{ a symmetric polynomial of degree } 2m_0 \text{ in the } x_i^{-1}, \tag{3}$$

hence (thesis lemma, `RamificationAfterBlowupIsTame.tex` lines 549–572) lies in $\kappa_C = k(e_1/e_d, \ldots, e_{d-1}/e_d)$ — i.e. has *no pole at* V_C — if and only if

$$\boxed{2m_0 < d, \quad \text{that is} \quad u_2 < d: \text{ the ramification bound.}}$$

Likewise $\sum_i a_1(x_i) \in \kappa_C$ iff $m_1 < d$, again the bound. Therefore, *under the bound*, both Witt components of $\vec{\alpha}$ lie in $\kappa_C \subset \kappa_C[[e_3]]$, the W_2-datum is integral, and the extension E/K_C defined by $\wp(\vec{Y}) = \vec{\alpha}$ is **unramified at** V_C.

Theorem 1. *For $p = 2$, $G = \mathbb{Z}/4$, $d = 3$ and $\mathcal{P}$ totally ramified at $\mathfrak{p}_\infty$ with ramification bounded by d (so $u_2 = 2 < 3$), $\mathcal{P}^{[3]_G}$ is unramified at η_C. The top Artin–Schreier class is not absorbed.*

Proof. $m_0 = 1$, $m_1 \le 2$, so $2m_0 = 2 < 3$ and $m_1 < 3$. By Section 3 both components of $\vec{\alpha}$ are in κ_C; explicitly $\alpha_0 = e_2/e_3$ and $\alpha_1 = \sum_i a_1(x_i) + e_1/e_3$, each of $v_C = 0$. An integral W_2-datum gives an unramified extension, so E/K_C is unramified at V_C. □

4 The boundary $d = r = 2$ reproduces the counterexample

The same formula (3) explains *why* the inductive route failed at the boundary. For $d = 2$ (so $r = d = 2$, $m_0 = 1$) the carry is

$$e_2(s) = s_1 s_2 = \frac{1}{x_1 x_2} = \frac{1}{e_2} \;\; (v_C = -1): \quad \text{a genuine pole.}$$

With $a_1 = 0$ the top component is $\alpha_1 = 1/e_2$, of break 1, so E/K_C is ramified — exactly the absorption/counterexample of **??**. The transition is sharp:

m_0	$u_2 = 2m_0$	d	carry at V_C	verdict
1	2	2	$1/e_2$ (pole)	**ramified** (counterexample)
1	2	3	e_1/e_3 (unit)	**unramified** (Theorem 1)
2	4	3	pole	ramified ($u_2 = 4 \not< 3$: bound fails)
2	4	5	unit	unramified

(All rows verified in `wittcheck.py`, step F.) The pattern is uniform: for $p = 2$, $r = 2$ the theorem holds *iff* the bound $u_2 < d$ holds, and the single Witt carry is the entire obstruction.

5 What this does and does not settle

- **Settled (rigorously, machine-checked):** $p = 2$, $r = 2$, all d. The theorem holds exactly under the bound; the obstruction is the Witt carry $e_2(x^{-m_0})$, integral iff $2m_0 < d$. The first case past the boundary, $(r, d) = (2, 3)$, is unramified — the class survives.
- **Conjecture for the mechanism.** For general p, r the datum of $\mathcal{P}^{[d]_G}$ is the length-r Witt sum $\sum_i^W \vec{a}(x_i)$; its higher components are symmetric polynomials built from the carry polynomials C (degree p), nested $r - 1$ times. Each such term is $e_k(x^{-m_\bullet})$-like and lies in κ_C iff its total "pole degree" is $< d$. The expectation is that these pole degrees are exactly the upper breaks $u_1, \ldots, u_r$, so the bound $u_r < d$ kills all carries simultaneously — the clean generalisation of Theorem 1.
- **Open.** (i) That the nested carries' pole degrees equal the u_j (a Schmid–Witt/Brylinski-type identity inside the symmetric power); (ii) $p \geq 3$, where C has $p - 1$ mixed terms rather than the single product $a_0 a_1 \mapsto a_0 a_0$ of $p = 2$. The first such test is $p = 3$, $r = 2$, $d = 4$ (the smallest genuine case there, since $u_2 \geq 3$).

Bottom line. The answer to "does $[a]$ survive at $(r, d) = (2, 3)$?" is *yes*, and not narrowly: for $p = 2, r = 2$ the wild Witt carry is the whole obstruction and the ramification bound is precisely what removes it. This is the direct, carry-level confirmation that the route killed for the wrong reasons in `a_wrong_path.tex` / `strategy_D_dead_end.tex` does close once the global bound is fed in — exactly as predicted.